

MULTIMODAL LONGITUDINAL MODELING OF PARKINSON'S DISEASE PROGRESSION

STATUS: PHASE 1 SYNTHESIS
DATE: SEPTEMBER 2026
COHORTS: PPMI & mPower

Comprehensive Literature Extraction, Critical Analysis, and Architectural Positioning

Rigorous extraction across all core reference papers, methodological strengths, unaddressed vulnerabilities, and capstone roadmap integration.

1. Executive Synthesis & Foundational Research Motivation

Parkinson's Disease (PD) is an exceptionally heterogeneous neurodegenerative disorder. Designing predictive models for disease progression requires navigating three complex axes: **temporal dynamics** (irregular clinical visits and non-linear decline), **multimodal integration** (combining motor exams, non-motor questionnaires, CSF/blood biofluids, and 3D neuroimaging), and **clinical confounding** (dopaminergic medication unmasking motor status). This review synthesizes six foundational papers to establish exact benchmark baselines, extract architectural principles, and identify fatal data-leakage and clinical gaps that our capstone project is uniquely engineered to resolve.

2. Literature Taxonomy: Three Distinct Scientific Streams

STREAM A: In-Clinic PPMI Multimodal Deep Learning	STREAM B: Real-World Ecological Smartphone Phenotyping	STREAM C: Distributed Big Data Machine Learning
• Dentamaro et al. (2024): 3D DenseNet + Excitation Network intermediate co-learning fusion on PPMI 3D MRI + Cardinal motor signs. • Junaid et al. (2025): Multitask BiLSTM + VAE + 3D CNN on PPMI time series to jointly predict Hoehn & Yahr stage and depression. Key Trait: Highly detailed clinical ground truth (MDS-UPDRS, DaTSCAN, MRI) collected under strict clinical trial protocols.	• Johnson & Logeswari (2026) / Botha et al.: Domain-Adaptive Transformer (DAT-PD) for continuous UPDRS Part II voice regression on mPower. • Bot et al. (2016): Foundational mPower study of 9,582 participants tracking intra-day medication responses via iPhone ResearchKit. Key Trait: High-frequency, naturalistic ecological monitoring; robust to microphone noise; captures daily motor fluctuations.	• Dritsas & Trigka (2025): Apache Spark MLlib benchmarking on massive datasets (1.4M to 100M rows) proving GBT superiority. • Chung / Lim et al. (2019/2025): Classifier-level stacking ensemble for multi-sensor wearable IMU fusion in daily living activities. Key Trait: Horizontal distributed computing and late decision ensembling for heterogeneous asynchronous sensor streams.

3. Alignment with Project Research Questions (RQ1 & RQ2)

Research Question 1 (RQ1): Trajectory vs. Snapshot Value	<i>Can disease progression at +12m and +24m be predicted more accurately from longitudinal visit sequences than from a static clinic snapshot?</i> • Literature Baseline: Junaid et al. demonstrated that multi-visit BiLSTMs improve stage classification as visits increase (from 2 to 6 steps). • Our Capstone Finding: At +12m, static snapshot and sequence models are virtually tied (MAE ~4.7-4.8 pts). However, at +24m, our Longitudinal BiLSTM sequence model decisively outperforms the static XGBoost snapshot (MAE 4.840 vs 4.971, R² 76.4% vs 75.1%), proving that multi-visit historical velocity is vital for long-range forecasting.
Research Question 2 (RQ2): Multimodal Fusion Architecture	<i>Which fusion strategy (Early, Late, or Cross-Modal Attention) yields the highest accuracy and temporal stability across clinical endpoints?</i> • Literature Baselines: Dentamaro et al. proved that intermediate cross-attention (Excitation Network) outperforms flat concatenation. Chung/Lim et al. proved that late decision ensembles handle sensor noise better than early feature stacking. • Our Capstone Roadmap: Direct head-to-head empirical comparison across Early Concatenation, Late Ensembling, and Cross-Modal Attention Fusion.

4. Master Literature Extraction Matrix (Official Proposal Template)

Standardized comparative extraction covering all 6 reference papers: Cohort & Sample Size, Exact Prediction Target, Fusion Strategy, Deep Learning / Classical Architecture, Split Protocol & Anti-Leakage Controls, Quantitative Results, and Critical Unhandled Realities.

Paper	Cohort / N	Exact Target	Fusion Strategy	Architecture	Split Protocol	Result	What They Didn't Handle
Dentamaro et al. (2024) <i>Sci. Reports (Nature)</i>	PPMI Cohort N = 90 subjects (Prodromal vs HC; baseline visit only)	Binary Class.: Prodromal PD vs. Healthy Control (HC) at baseline	Intermediate / Co-Learning: Excitation Network (EN) cross-conditions clinical features with 3D MRI latent feature maps	3D DenseNet-121 (MRI visual backbone) + Excitation Network (co-learning fusion) + Linear head. XAI: 3D Integrated Gradients	Stratified 10-fold CV across the 90 patients	Accuracy: 96.6% AUC: 99.4% F1-macro: 96.5% (MRI alone: 92.9% F1; Clinical XGB alone: 78.4% F1)	• Zero longitudinal tracking (only 1 baseline visit; no progression). • Tiny sample size (N=90; discarded >98% of PPMI). • Replaced missing entries with -1. • No medication (LEDD), DaTSCAN SBR, or CSF biofluids.
Junaid et al. (2025) <i>IEEE Access</i>	PPMI Cohort N = 1,059 pats (Time series up to 6 visits + baseline 3D T1 MRI)	Multi-Task Class.: 1. Hoehn & Yahr (NHY) stage (1–4) 2. Binary depression 3. 4-class depression	Hybrid Early + Late: Early tabular fusion compressed via VAE; fused late with 3D CNN MRI features into a BiLSTM	3D CNN (baseline MRI) + Variational Autoencoder (VAE) + Bi-LSTM (temporal) + Bayesian Opt + Multi-task heads	5-fold CV on 70% Train / 30% Test with oversampling	NHY Test Acc: 92.97% (CV: 96.78%, F1: 94.47%) at 6 steps. Depression Acc: 93.66%. (Multimodal beat unimodal 88.7% & 92.0%)	• No continuous score forecasting (only discrete integer stages 1–4). • Rigid fixed visit steps (2, 4, 6 visits), ignoring irregular delta_t days. • Static baseline MRI only. • Omitted DaTSCAN SPECT, CSF/SAA, and medication (LEDD).
Johnson & Logeswari (2026) <i>Front. Digit. Health</i>	mPower Cohort N = 5,800 pats (58,247 voice files; 3,088 PD train, 662 PD test)	Continuous Regr.: Longitudinal self-reported MDS-UPDRS Part II (0–52) over 24m trajectory	Domain-Adaptive Acoustic Fusion: Fuses 88 eGeMAPS acoustic features with MFCCs; aligns microphone shifts via DANN / CORAL	DAT-PD: Acoustic preprocessor -> eGeMAPS/MFCC -> Domain-Adaptive Transformer -> GRU Trajectory Decoder -> Regr. Head	Strict Patient-Level Split (Train: 4,060, Test: 662; 0% overlap) with 10-fold CV	MAE: 2.74 pts (95% CI: 2.44–3.01) RMSE: 3.61 pts R²: 0.93 Pearson r: 0.965 (Beat LSTM MAE 3.85, SVM MAE 6.82)	• Self-reported Part II questionnaire, NOT clinical Part III motor exam. • Voice only (no gait, tapping, or clinical labs). • Unverified app-based self-selection diagnosis. • Sustained vowel only; no conversational speech.
Bot et al. (2016) <i>Nat. Sci. Data (Foundational)</i>	mPower Cohort N = 9,582 pats (1,087 PD, 8,495 controls; iPhone ResearchKit)	Data Descriptor & Feasibility: Adherence, intra-day Levodopa response curves, and cross-sectional PD vs HC	Multi-Activity Sensor Suite: Independent sensor streams for voice, tapping, walking/gait, and memory; no unified fusion	Apple ResearchKit mobile pipeline -> Synapse cloud -> signal extraction (accelerometer, gyro, audio WAV) -> linear mixed models	Open Science Benchmark: Participant-level IDs (healthCode) for researcher partitioning	Feasibility demonstrated at massive scale (60k+ voice & tapping tasks); identified measurable intra-day motor fluctuation before vs after medication	• Dataset descriptor; no predictive progression modeling. • Extreme attrition (median retention was only 5–7 days). • Self-reported diagnosis without clinical validation. • Zero biofluids, imaging, or MDS-UPDRS Part III.
Chung / Lim et al. (2019/2025) <i>Sensors (MDPI)</i>	Wearable IMU N = 5 volunteers (8 body-worn IMUs: wrists, ankles, waist; 9 daily activity classes)	Multi-Class Class.: Recognition of 9 Activities of Daily Living (ADLs: walking, eating, driving, etc.)	Two-Level Classifier Ensemble (Late Fusion): Separate LSTMs per sensor, fused via accuracy-weighted soft voting & stacking	2-layer LSTM base learners (128 units) per sensor -> Stacking Ensemble with activity-customized decision weights	10-fold CV and Leave-One-Subject-Out (LOSO) validation	Ensemble Acc: 93.07% F1-score: 0.93 (Waist-alone baseline: 75.8% accuracy)	• Evaluated healthy volunteers for HAR, NOT Parkinson's patients. • Tiny sample size (N=5). • Instantaneous activity detection; no disease trajectory. • Heavy physical burden (8 body-worn sensors).
Dritsas & Trigka (2025) <i>IEEE Access</i>	Big Data Benchmark • NYC Taxi: 1.46M • Netflix: 100.5M • Higgs: 11.0M	Big Data Benchmarks: • NYC: Trip duration (MAE) • Netflix: Movie rating (RMSE) • Higgs: Signal detection (AUC)	Distributed Computing: Horizontal compute scaling via Apache Spark Catalyst optimizer & MLlib; not multimodal fusion	Apache Spark MLlib: Distributed GBT, Random Forest, LinR/LR, SVM, KNN across multi-node Spark clusters	80% Train / 20% Test split with 5-fold CV on training set	Near-linear execution scaling up to 100M rows. GBT won all tasks: • Taxi: RMSE 0.38, MAE 0.28 • Netflix: RMSE 0.88, MAE 0.69 • Higgs: AUC 0.824, Acc 74.2%	• Industrial benchmark domains; zero medical/clinical relevance. • Did not handle patient clustering or irregular longitudinal clinic visits. • Flat tabular data; no multimodal alignment or deep feature fusion.

5. In-Depth Methodological Strengths & Critical Weaknesses

Detailed examination of core innovations, biological validities, algorithmic strengths, and fatal vulnerabilities for each study.

1. Dentamaro et al. (2024) — 3D DenseNet + Excitation Network Co-Learning KEY METHODOLOGICAL STRENGTHS: • [-] Demonstrates that intermediate co-learning fusion outperforms simple feature concatenation. • [+] 3D Integrated Gradients provides voxel-level attribution maps confirming substantia nigra focus. • [+] Proves clinical features boost deep MRI visual representation (+3.6% to +26% F1). CRITICAL FLAWS & UNADDRESSED GAPS: • [-] Severe cohort attrition: filtered PPMI down to only 90 subjects (high overfitting risk). • [-] Cross-sectional only: zero longitudinal progression modeling or future forecasting. • [-] Crude imputation: replaced all missing tabular entries with -1. CAPSTONE TAKEAWAY: Validates intermediate cross-attention for RQ2, but highlights the necessity of scaling to our full cohort of 5,426 patients.	2. Junaid et al. (2025) — Multitask BiLSTM + VAE Sequence Modeling KEY METHODOLOGICAL STRENGTHS: • [+] Formulates multitask learning coupling non-motor depression with motor staging. • [+] Variational Autoencoder (VAE) compresses noisy clinical time series effectively. • [+] Multimodal integration significantly beats monomodal baselines (92.97% test accuracy). CRITICAL FLAWS & UNADDRESSED GAPS: • [-] Discretizes severity into coarse integer stages (NHY 1-4) rather than continuous motor scores. • [-] Assumes rigid fixed visit steps (2, 4, 6), ignoring real-world irregular visit intervals. • [-] Omitted DaTSCAN SPECT, CSF biofluids, and active Levodopa medication confounding (LEDD). CAPSTONE TAKEAWAY: Directly inspires our Model 2 BiLSTM architecture while underscoring the need to model irregular delta_t and continuous progression.
3. Johnson & Logeswari (2026) / Botha et al. — DAT-PD Domain-Adaptive Voice Transformer KEY METHODOLOGICAL STRENGTHS: • [+] True continuous longitudinal trajectory forecasting ($R^2 = 0.93$, Pearson $r = 0.965$). • [+] Domain-adversarial training (DANN/CORAL) successfully removes smartphone microphone noise. • [+] Strict participant-level zero-leakage split across 5,800 participants. CRITICAL FLAWS & UNADDRESSED GAPS: • [-] Predicts subjective self-reported Part II survey, NOT clinician-administered Part III motor exam. • [-] Voice-only modality cannot capture lower-limb gait freezing or postural instability. • [-] Unverified smartphone app downloads introduce diagnostic noise. CAPSTONE TAKEAWAY: Establishes the state of the art for continuous trajectory regression and validates strict participant-level splitting.	4. Bot et al. (2016) — Foundational mPower Study & ResearchKit KEY METHODOLOGICAL STRENGTHS: • [+] Landmark open-science dataset establishing digital biomarker feasibility at scale (N=9,582). • [+] Captured real-world intra-day motor symptom fluctuations before vs after taking Levodopa. • [+] Multi-activity suite (voice, finger tapping, 30-step walk, memory). CRITICAL FLAWS & UNADDRESSED GAPS: • [-] Dataset descriptor paper; did not build a predictive disease progression model. • [-] Severe attrition: median participant engagement was only 5–7 days. • [-] Younger, tech-savvy demographic skew with zero in-clinic neurological exams. CAPSTONE TAKEAWAY: Highlights the paramount clinical importance of controlling for medication timing (hr_post_med, pd_state_on/off) in modeling.
5. Chung / Lim et al. (2019/2025) — Two-Level Classifier-Level Sensor Ensemble KEY METHODOLOGICAL STRENGTHS: • [+] Systematically proves that late decision-level voting beats early feature concatenation for sensors. • [+] Empirically optimizes sensor placement (bilateral wrists, ankle, waist) and sampling rate (10 Hz). • [+] Activity-specific customized decision weights maximize multi-modal consensus. CRITICAL FLAWS & UNADDRESSED GAPS: • [-] Evaluated healthy volunteers for HAR, NOT Parkinson's patients. • [-] Microscopic sample size (N=5 volunteers). • [-] High hardware burden (8 body-worn sensors) impractical for elderly clinical populations. CAPSTONE TAKEAWAY: Provides architectural foundation for Late Decision Fusion benchmarking in Phase 4 (RQ2).	6. Dritsas & Trigka (2025) — Apache Spark MLlib Distributed Computing KEY METHODOLOGICAL STRENGTHS: • [+] Rigorous empirical benchmark of Apache Spark scalability across 100M+ records. • [+] Proves Gradient-Boosted Trees (GBT) consistently outperforms RF and linear models. • [+] Provides hardware profiling (CPU, RAM, latency per data partition). CRITICAL FLAWS & UNADDRESSED GAPS: • [-] Evaluated industrial non-medical domains (NYC Taxi, Netflix ratings, Higgs Boson). • [-] Zero healthcare, clinical time series, or patient-level clustering relevance. • [-] Flat tabular data only; no multimodal alignment or deep feature fusion. CAPSTONE TAKEAWAY: Directly justifies our Phase 2 Apache Spark data engineering pipeline and our Model 1 XGBoost baseline.

6. Strategic Positioning: How Our Pipeline Bridges Every Literature Gap

Direct side-by-side gap comparison demonstrating how our capstone architecture resolves the clinical, computational, and methodological shortcomings identified across the published literature.

Literature Vulnerability / Unaddressed Reality	Affected Papers	How Our Capstone Pipeline Resolves It	Clinical & Scientific Significance
Severe Sample Size Restriction Filtering cohort to small homogeneous subsets	Dentamaro et al. (N=90) Chung/Lim et al. (N=5)	Full Cohort Ingestion: 5,426 patients across 32,743 visits (1,829 PD, 340 HC, 3,176 Prodromal, 81 SWEDD).	Eliminates severe overfitting; captures realistic population variance and prodromal transition dynamics.
Discrete vs. Continuous Progression Predicting coarse integer stages (NHY 1-4) or binary detection only	Dentamaro et al. (Binary) Junaid et al. (NHY 1-4)	Continuous Progression Forecasting: Direct regression of MDS-UPDRS Part III continuous motor score (0–132 pts) at +12m and +24m, alongside MCID (>=3.5 pts) rapid worsening classification.	Aligns with clinical trial endpoints; provides actionable continuous sensitivity required for disease-modifying therapies.
Subjective Self-Report vs. Clinical Exam Relying on unverified smartphone questionnaire surveys	Johnson & Logeswari (Part II) Bot et al. (App self-reports)	In-Person Neurological Gold Standard: Clinician-administered MDS-UPDRS Part III motor exams, supplemented by DaTSCAN SPECT striatal binding ratios and CSF biofluids.	Ground truth confirmed by board-certified movement disorder specialists; immune to self-reporting bias.
Rigid Fixed-Step Time Assumptions Assuming uniform intervals (steps 2, 4, 6) and ignoring missing visits	Junaid et al. Dentamaro et al.	Irregular Spacing & Temporal Interval Awareness: Explicitly ingests inter-visit intervals (delta_t_days) and cumulative elapsed months from baseline into recurrent sequence encoders.	Reflects real-world clinical follow-up; accurately models progression velocity irrespective of appointment spacing.
Medication Confounder Fluctuation Ignoring Levodopa dose and motor testing state (ON vs OFF drug)	Junaid et al. Dentamaro et al. Dritsas & Trigka	Active Confounder Controls: Explicitly engineers active Levodopa Equivalent Daily Dose (LEDD mg), testing state (pd_state_on/off), and hours post-medication (hr_post_med).	Prevents drug symptom masking from being misclassified as true disease stabilization or recovery.
Risk of Patient Data Leakage Random row splitting across longitudinal visit trajectories	Common pitfall in literature (Dritsas et al. standard random splits)	Strict Patient-Level Partitioning: All splits partitioned strictly by PATNO (Train 70% = 3,787 pats, Val 15% = 827 pats, Test 15% = 812 pats; 0% patient overlap).	Guarantees 100% zero data leakage; ensures test results reflect true generalization to unseen clinical patients.

7. Integration into Project Execution Roadmap

Phase 3 Modeling	Model 1 (Completed): Static XGBoost Snapshot (MAE = 4.711 pts, R² = 0.7740). Model 2 (Completed): Longitudinal BiLSTM Trajectory with Temporal Attention (MAE = 4.840 pts, R² = 0.7636 at +24m). Outperforms Model 1 on 2-year forecasting, directly answering RQ1. Model 3 (Upcoming): Monomodal Neuroimaging Baseline (DaTSCAN SPECT SBR + FreeSurfer MRI subcortical volumes alone).
Phase 4 Multimodal Deep Fusion	- Directly implements and benchmarks the three classic fusion paradigms to resolve Research Question 2 (RQ2): <i>Early Feature Concatenation</i> (merging multimodal latent vectors pre-sequence). <i>Late Decision Ensembling</i> (stacking/voting inspired by Chung/Lim et al.). <i>Cross-Modal Attention Fusion</i> (intermediate co-learning inspired by Dentamaro et al.'s Excitation Network).